

A Framework for Emerging Risks



BY JOSEPH A. IRACI AND CHAD ROGERS

RISK MANAGERS NEED to be aware of emerging risks—not only because they can damage and even eliminate firms and industries, but also because they can present business opportunities for those who capitalize on them.

As evidence of the former, consider IBM, one of the most successful firms in history. It didn't see the market potential in software, though, giving rise to Microsoft. For their part, U.S. automobile and motorcycle manufacturers failed to develop a strategy to defend market share against smaller, high-quality, and low-cost vehicles from Japan. And just this summer we witnessed the demise of Toys 'R' Us, which fell short of developing a successful e-commerce strategy.

Conversely, Western Union, founded in 1851, transformed itself from a sender of telegrams into a communications and money-transfer company. And National Geographic—which printed its first magazine in 1888—found a new identity through a successful multimedia strategy.

In today's interconnected world, with its ever-faster technological advances, risks are emerging at an alarming rate. And our traditional ways of managing them—heavily weighted toward quantitative models—are not sufficient to address their scale, speed, or considerable tail risk. More qualitative analysis is required.

The threats are indeed existential. CNBC noted this year that 86% of the companies that made the original Fortune 500 list in 1955 no longer exist. Many of the fallen simply did not adjust to changing circumstances that brought significant risks to their business model.

Going forward, it is reasonable to think that surviving will not get any easier. Today's fast-paced, data-laden operating environment requires risk managers to use an integrated framework for identifying, assessing, and managing emerging risks. Successful firms need to be learning organizations built on intellectual capital. They must have operating models that allow for flexibility to both adjust to changing circumstances and capitalize on new opportunities. Indeed, risk represents not only downside exposure but also upside potential.

What Is Emerging Risk?

Risk can be defined as the potential for deviation from an expected result. This deviation can be positive or negative. But even a positive outcome requires analysis to understand what drove the deviation. For example, was it driven by skill or sheer luck?

An emerging risk is a risk that, if it manifests, could preclude a firm from meeting its business objectives. Examples of emerging risks in today's environment include global warming, cyber risk, and artificial intelligence.

Think of the relationship between risk and emerging risk in terms of living next to a volcano: People know there is a theoretical risk, but they do not know when the emerging risk—an eruption—will be realized.

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A Framework for Emerging Risks

Risk Culture: Risk Appetite, Liquidity, and Capital

Risk appetite is an organization's articulation of how much risk it is willing to assume. It translates risk metrics and methods into thresholds or tolerance levels that directly influence business decisions. It is the bridge that connects corporate strategy with risk management.

Linking risk appetite to corporate strategy allows management to constantly assess how emerging risks might affect the completion of objectives, and it ensures that the appropriate level of capital is deployed to support the firm's strategy at any given time. Liquidity and capital management play a critical role in ensuring firm solvency, especially during stressful periods like the 2008 recession. (See sidebar, "Liquidity and Capital Management: A Primer.")

Liquidity measures a firm's ability to meet its obligations. These obligations can be met by using cash on hand, converting assets to cash, or accessing lines of credit. During normal times, liquidity functions smoothly. But in times of extreme stress, liquidity can dry up quickly. Firms must prepare for the possibility of not having access to liquidity during stressed conditions. If this situation is handled incorrectly, an already tense situation will simply become worse.

Capital is a financial cushion needed to absorb unexpected losses and support business growth. Expected losses are those the firm expects to incur and are considered a cost of doing business. Unexpected losses are those that exceed the amount of expected loss. Unlike liquidity, capital can take several forms, ranging from economic capital to regulatory capital. Capital is sometimes referred to as the difference between assets and liabilities on the balance sheet.

Regulatory capital is typically comprised of market risk, credit risk, and operational risk. In contrast, economic capital usually has more inputs and can include liquidity risk, business risk, interest rate risk, market risk, credit risk, and operational risk.¹

Economic capital is based on the probability of potential future losses and is a forward-looking measure of capital. The primary value of economic capital is its ability to influence decision making. Its scale of influence depends on the development of an allocation method that best suits the firm and the establishment of risk-adjusted return on capital, hurdle rates, and target rates.

Risk Governance and Processes

Developing a risk governance process for emerging risks will allow for a repeatable process and position the firm for success. This overall process, however, cannot be restricted to risk management; it must include subject-matter experts from within the business and across corporate functions. Including people from different backgrounds and disciplines is recommended because it can counter the impact of possible cognitive bias in the group. Moreover, people with different backgrounds and skill sets can also make for a richer scenario analysis by providing a wider scope of assumptions and discussions.

Identifying and Assessing Emerging Risks

By being aware of business objectives and fostering a link between risk appetite and strategy risk, managers are better equipped to deploy effective monitoring systems that cover the external and internal environments. The data flows into risk systems that try to identify volatility catalysts, which could come from within or outside the firm. The volatility catalysts are indicative of emerging risks. And when these risks manifest themselves, the probability for tail-end events increases.

BY BEING AWARE OF BUSINESS OBJECTIVES AND FOSTERING A LINK BETWEEN RISK APPETITE AND STRATEGY RISK, MANAGERS ARE BETTER EQUIPPED TO DEPLOY EFFECTIVE MONITORING SYSTEMS THAT COVER THE EXTERNAL AND INTERNAL ENVIRONMENTS.

But risk managers cannot rely on quantitative models alone to help determine when risks will be realized. The following are examples of emerging risks that do not lend themselves to modeling but must be addressed in order to provide the qualitative analysis for proper positioning of the firm.

A Possible Misstep by Central Banks

- Intervention by central banks during and after the financial crisis was unprecedented. Now the world's central banks are returning to a normalized monetary policy, and no one knows for sure what the impact will be.
- There is a risk the Federal Reserve will raise interest rates too quickly and slow the economy. Conversely, rates rising too slowly, combined

with rapid economic growth fueled by tax cuts, could spur inflation.

- China is attempting to rein in excessive debt while the government enacts reforms aimed at transforming the economy into a more service-oriented model.

Continued Threats to the EU

- Brexit concerns, political instability in Italy, uncertainty in Hungary, and clashes over immigration all have the potential to add stress on the EU. The general lack of wage flexibility and fiscal payments between EU member countries further compounds this issue.

Geopolitical Issues

- Tension in the Middle East and fear of escalating conflict between the West, Russia, and Iran are likely factors in the run-up of energy prices.

Trade Tensions

- The trade spat with China continues to evolve rapidly as markets react sharply to new headlines. Additionally, the effect of tariffs may lead to ineffective production and deadweight loss.

The above risks are systemic and measured by the beta coefficient. Beta is the amount of systemic risk—also known as volatility—a security or portfolio has relative to the overall market. The overall market has a beta of 1, and all assets are measured against this baseline. So if a security or portfolio has a beta greater than 1, it is more volatile than the overall market. If the beta is less than 1, the security or portfolio is less volatile than the market.

The flip side of systemic risk is idiosyncratic risk, which is specific to a company or asset class and has no correlation to broad market moves. Systemic risk cannot be diversified, but idiosyncratic risk can be mitigated by diversification.

Common drivers of systemic risk:

1. Monetary policy
2. Fiscal policy
3. Inflation
4. Growth

Common drivers of idiosyncratic risk:

1. Management
2. Strength of financials
3. Culture
4. Location
5. Industry

Risk managers need to be aware of systemic and idiosyncratic risks, as either or both can drive emerging risks. It is the combination of both that represents total risk.

Data

Data is a critical component underpinning risk management. Data falls into three categories: 1) what is known, 2) what is not known but should be known, and 3) things that will never be known. For known data points, risk managers must identify known risks and assess their potential impact. This is the first step in identifying emerging risks and their associated effects, which must be analyzed and risk-graded before responses to them are drafted.

There is other information that risk and business managers do not know but should know. Such gaps must be closed because they represent potential emerging risk. For example, a trader is providing the mark-to-market on his or her own trading position, or a single individual has access to both front- and back-office systems. Not knowing what should be known has engulfed many a firm in crisis.

That leaves certain risks that will never be known with certainty. Within this data classification are emerging risks and random events. It is often difficult to predict when emerging risks will manifest themselves. But it is nearly impossible to predict when random events, such as earthquakes, transit strikes, and flu epidemics, will

LIQUIDITY AND CAPITAL MANAGEMENT: A PRIMER

BY JOSEPH IRACI

LIQUIDITY AND CAPITAL management play a critical role in ensuring firm solvency, especially in stress scenarios like the 2008 recession, and as a bulwark against emerging risks (for more on these risks, see the main article).

The terms, liquidity and capital, are often used interchangeably, but they are distinct and serve different purposes in ensuring firm solvency. Firms have become insolvent because of inadequate capital and/or liquidity, not only causing loss of shareholder value but shaking confidence in the stability of the financial system.

Liquidity

Liquidity measures a firm's ability to meet its obligations. These obligations can be met by using cash on hand, converting assets to cash, or accessing lines of credit. A firm's liquidity refers to how quickly it can convert its mix of assets to cash or access other sources of funds to meet obligations. During normal times, liquidity functions smoothly. But during periods of extreme stress, liquidity can dry up quickly and exacerbate an already tense situation.

It is during these stressed times that funding risk and market risk are at their highest levels. Funding risk refers to a firm's ability to meet its obligations through available cash or collateral. Market risk refers to the ease with which assets can be sold in the market to raise liquidity to meet firm obligations.

Firms can mitigate these two risks in various ways. One way is to ensure access to stable funding sources that hold up even under stress scenarios. Bank regulators call this type of funding—demand deposits and statement saving accounts—a bank's "core deposits." The point, even for nonbanks, is to evaluate the stability of funding sources

in multiple scenarios. Overreliance on more volatile sources of borrowing leaves firms vulnerable in periods of stress.

Another mitigation is to hold a buffer of liquid assets and cash that can be relied on to meet obligations. As part of liquidity planning, it is important for firms to periodically review existing liquidity strategies and activities, identify primary and secondary sources of funding, and ensure liquidity management that is consistent with the board's risk tolerance.

As seen during the 2008 recession, even instruments once thought of as liquid may not be if the situation is extreme. A number of firms had funding and liquidity problems in 2008 because normally liquid assets were difficult to sell, and even products thought of as cash equivalents did not have markets.

Firms need to be prepared for the risk of not having access to liquidity during a stressed environment. Coming out of the 2008 recession, the Basel Committee agreed on a liquidity coverage ratio (LCR) and a net stable funding ratio (NSFR). The LCR addresses a firm's ability to maintain a liquidity buffer during a short-term stress scenario, while the NSFR addresses stable funding structures. In addition to these ratios, firms need to develop the ability to stress test liquidity in hypothetical scenarios and consider the various sources of liquidity and the risk posed by illiquidity.

Capital

Capital is a financial cushion needed to absorb unexpected losses and support business growth. Expected loss is an estimate of losses a firm expects to incur and are considered a cost of doing business. Unexpected losses are those losses that exceed the amount of expected loss.

The best example of how expected loss and unexpected loss work is a loan portfolio. Based on the credit quality of a

occur. Even so, they still need to be addressed through business continuity and disaster recovery planning, scenario analysis and simulation exercises, and insurance coverage. Known or emerging risks must be analyzed to assess the likelihood, severity, and speed with which they will surface. If the risk presents a material exposure to the firm, then it must be escalated. But the speed of random events is always quick, and the severity of impact must be assessed quickly and escalated if necessary.

Whether an event is driven from an emerging risk or a random occurrence, the key is to have a coordinated response based on set criteria and escalation protocols that involve the appropriate level of management. A white paper by Kathleen Locklear² notes that, for emerging risks, management should reduce their focus on predictive models in favor of alternative approaches that could allow emerging risks to be anticipated more accurately.

Locklear also details the impact that cognitive impediments have on decision making, saying they could preclude management from acting because “biases may produce a distorting effect, one which helps to explain why early warning signals are inexplicably discounted and/or missed ...during the incubation period that precedes disasters.”

That statement implies a difference between event management and crisis management, and it is true that the vast majority of risk events never rise to the level of a crisis and are handled as part of the daily routine. Ideally, there is a systematic way to capture and identify data and inform executive management of those events that could preclude the firm from meeting its business objectives. Studies have shown that executive management's engagement is critical to minimizing negative impacts on a firm's reputation and avoiding dire consequences in a crisis.

Behavioral Impediments and Risk Mitigation

The economic graveyard is littered with companies that either failed to identify and act upon changes or saw changes but decided not to act. This inability to capitalize on emerging risks cuts across the globe and all industries. As pointed out in Locklear's white paper, the existence of behavioral impediments distorts decision making.

Cognitive bias refers to the reliance on limited information or preconceived notions when making decisions. It often causes people to make illogical decisions based on faulty or insufficient information or on previously held beliefs. Meanwhile, emotion refers to our frame of mind and is often described as our mood. As an example, the uncertainty that fear introduces could cause an illogical decision.

In addition to simple analysis of numbers, there are individual behaviors, emotions, personality traits, and life experiences that influence our decision making. Risk managers use a combination of quantitative and qualitative analyses, but all decisions ultimately are judgment calls.

There are hundreds of known biases. Below are several cognitive ones that risk managers need to know about. They should also be aware that biases do not exist in isolation but rather interact with one another.

- **Confirmation bias:** Refers to the behavior where people seek information that confirms a preexisting belief, ignoring more objective information.
- **Anchoring:** A bias where a person may jump to a conclusion based on the first piece of information he or she receives, or possibly hold on to a belief and jump to a conclusion.
- **Overconfidence:** A result of people having too much faith in their own analyses or abilities—possibly leading, in the case of the financial industry, to higher-risk asset purchases or concentrated positions.

Regulatory capital is typically comprised of market risk, credit risk, and operational risk. In contrast, economic capital usually has more inputs and can be comprised of liquidity risk, business risk, interest rate risk, market risk, credit risk, and operational risk.

Source: FDIC, *Supervisory Insights*, “Economic Capital and the Assessment of Capital Adequacy.”

loan portfolio, a bank can estimate its credit losses over a given period of time. The bank would establish a loan loss reserve through a provision expense on the income statement, and the credit would reside as a contra-asset on the balance sheet. The loan loss reserve is the expected loss of the current portfolio. But in an extreme-stress scenario, credit quality may deteriorate so quickly that capital is relied on as a buffer for unexpected loss.

Capital is also used to support the growth of the business. Capital on the balance sheet is the difference between assets and liabilities. Forms of capital include economic capital and regulatory capital.

Economic capital is a measure of risk and distinct from regulatory capital, which is the amount of capital required by regulators. Economic capital is based on the probability of potential future losses. It is a forward-looking measure that is usually considered the difference between a given percentile of a loss distribution and the expected loss.

A fully developed economic capital model has the potential to better equip a firm to anticipate potential issues. The

primary value of economic capital is its ability to influence decision making and risk management. But its influence on decision making depends on the development of an allocation method that best suits the requirements of the firm and the establishment of a risk-adjusted return on capital, hurdle rates, and target rates.

- **Allocation:** Firms need to determine the method of allocation that best suits them. It stands to reason that allocation should begin at the business unit or division level, with more detailed allocations made at the discretion of management. Beyond the business unit or divisional allocation, economic capital should be allocated to each of the main drivers of risk. These drivers of risk, in turn, will depend on the business model of each specific firm.
- **Risk-Adjusted Return on Capital (RAROC):** RAROC is simply expected profit divided by economic capital, the expected profit being revenue minus expenses.
- **Hurdle Rates and Target Rates:** Hurdle rates establish a minimum rate of return needed. Each project or product must meet the minimum return, and the hurdle rate is determined based on the cost of capital for the firm. Target rates, in comparison, establish the rate of return that management is setting its sights on.

Economic capital and regulatory capital both play critical roles in capital management, but the nature of economic capital makes it especially well-suited for capital planning. Complementary to the calculation of capital is the incorporation of stress testing, where firms deploy forward-looking stress tests that evaluate a firm's ability to absorb losses and continue operations in a stress environment.

Capital Plans and Stress Tests

The Comprehensive Capital Analysis and Review (CCAR) is used by the bank regulators to assess the capital adequacy of banks. As of 2014, even mid-market

banks (those having assets of \$10-50 billion) are required by the Dodd-Frank Act to submit capital stress tests to the regulators.

Banks with assets greater than \$50 billion have been submitting plans since 2011 and face more requirements. However, many industry watchers expect CCAR requirements to be relaxed somewhat given the deregulation drive in Washington, D.C., particularly since other requirements have been curtailed through passage of the Economic Growth, Regulatory Relief, and Consumer Protection Act.

The recent legislation, for example, frees banks with up to \$100 billion in assets (the number will ultimately rise to \$250 billion) from Dodd-Frank Act stress tests (DFAST). DFAST requires banks to assess their capital position under baseline scenarios developed by the regulators. Banks take different variables—GDP changes, unemployment spikes, market crashes—and use their internal forecasting models to determine the effects on revenue, losses, liquidity, and capital.

Currently, CCAR requires banks with assets of more than \$50 billion to develop and submit capital plans to the regulators, who use these submissions to determine if the bank is able to maintain minimum regulatory capital under scenarios of extreme stress. A 2012 article from the Federal Reserve Bank of New York, "CCAR: More Than a Stress Test," noted that "an important feature of capital plans is that they are explicitly forward-looking. The plans focus not just on the BHC's current regulatory capital ratios, but also on assessments of what might happen to its capital in the future under both expected and stressful economic conditions. This forward-looking assessment is important because regulatory capital ratios do not always adjust rapidly as market conditions and perceptions about the firm change, especially during times of stress." The same article also detailed required elements of the capital plan:

- A detailed description of the overall process.

- The guidelines, principles, and governance procedures for the capital policy.
- A description of planned capital actions.
- An assessment of sources and uses of capital over a nine-quarter horizon.

Liquidity and capital are related but also distinct from each other. Liquidity is related to sources and uses of funds in order to meet firm obligations. Capital, in contrast, is a buffer to absorb losses. Either liquidity or capital could have major adverse implications for firms, and in extreme situations both can have problems simultaneously.

Accordingly, regulators have increased their expectations for bank managements to be proactive in assessing the impacts of liquidity and capital during periods of extreme stress. The objective is to ensure that firms maintain solvency and that they can continue to play a critical intermediary role in the global economy.

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- **Planning fallacy:** Occurs when a person underestimates the time needed to complete a task or overrates the ability to shape the future.
- **Choice paralysis:** Happens when too many choices or too much information leads to the inability to make a decision.
- **Loss aversion:** Refers to the tendency for people to sell assets that have a gain while retaining assets that have a loss. Research has shown that people simply feel the pain of a loss more than the joy of a gain.
- **Representativeness:** Occurs when we think recent events will continue into the future.
- **Herding:** Refers to following the crowd.

Think of past risk events and the motivations that led people into behavioral patterns that ultimately were destructive to themselves, the firms they worked for, and the industry. What motivated otherwise intelligent people to act in a way that was so harmful to so many stakeholders? Many common emotions are part of our makeup and have the potential to impact decisions. They include happiness, pride, confidence, concern, jealousy, resentment, insecurity, entitlement, fear, anger, and sadness.

In *Forbes*, Erik Larson wrote that recent research suggests emotional intelligence is a strong predictor of success at work: “When it comes to decision making, a little emotion is good, even if the emotions seem unpleasant or unproductive. Feeling a little fear, sadness, or irritation can help spark motivation or broaden the search for alternatives.”³ He warned, however, that problems could arise when there is too much emotion.

Logic and reason are important, but risk managers need to understand the emotional drivers that impact a person’s frame of mind when making decisions. To mitigate the negative influence of cognitive bias and emotion on decisions, it is important to be

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aware that both exist—and to have a thoughtful process that reduces their influence. Several approaches can be taken:

- **Self-awareness:** Know yourself and your frame of mind when making critical decisions and know the frame of mind of the decision makers in your organization.
- **Bias awareness:** Biases exist, but by being aware of them we can become better critical thinkers and decision makers. We can even mitigate against our own blind spots.
- **Situational awareness:** This refers to the ability to identify, process, and understand the critical elements of

information so that we know what’s happening around us. See things as they are, not as you wish them to be.

- **Objective:** This refers to knowing what you are trying to accomplish, which guides downstream decision-making during the execution phase.
- **Constructive conflict:** This requires people to focus on the end result and have an open discussion for the benefit of the overall team.
- **Different information sources:** Look for information from a wide range of sources to consider multiple perspectives.
- **Devil’s advocate:** Seek out people with different opinions to challenge perceptions and conclusions.
- **Brainstorming:** Have an open and free-wheeling discussion to generate ideas and challenge assumptions.

Enterprise Risk Management (ERM)

A fully integrated ERM program is one in which each risk discipline works within a dynamic process of gathering internal and external data that feeds into their respective risk systems. These separate data streams then become aggregated, taking into account intra-risk relationships.

The output of this exercise is information that can be used at the appropriate management level to ensure each has sufficient information to deal with uncertainty and support decision making. The value is that risk-sensitive information becomes an integral part of strategic and tactical decision-making and can mitigate the impact of emerging risks.

ERM requires a portfolio view of risk that provides a holistic take on the organization and seeks to understand how all risks can ultimately impact it.

Conclusion

As globalization and technology have made the marketplace much more competitive, it has become increasingly important for firms to be able to understand and plan for emerging risks—which can manifest quickly

and have the potential to disrupt any firm or industry.

Successful companies need to ensure they have the necessary financial and operational flexibility to adapt and take advantage of new opportunities. This requires risk managers to deploy data management and analytics that monitor the internal and external environments. The result is the ability to identify volatility catalysts and have plans in place to respond to them.

Quantitative modeling by itself is not enough to monitor emerging risks. It needs to be combined with qualitative analysis that monitors the environment and identifies when volatility catalysts have a higher probability of occurring. This approach requires an enterprise-wide view of risk manage-

ment and the adoption of a cross-discipline framework in which all the risk disciplines work in unison to provide critical information to management. 

Notes

1. See Federal Deposit Insurance Corporation, "Economic Capital and the Assessment of Capital Adequacy," *Supervisory Insights*, Winter 2004. Available at www.fdic.gov/regulations/examinations/supervisory/insights/siwin04/siwinter04-article1.pdf.
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3. See Erik Larson, "How the Most Common Emotions Affect Business Decision Making and What to Do About It," *Forbes*, March 21, 2017.



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